Value of Forecasting Accuracy - Day-Ahead Auction Only

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1 Introduction

1 Overview

This document shows the mathematical formulation of a comparison of the energy arbitrage margins of a Battery Energy Storage System (BESS) in the Day-Ahead Auction (DAA) power market in Germany.

The analysis compares trading margins of optimizing trading decisions based on different price forecasts.

The optimization uses a BESS dispatch optimization over quarter-hours ($q \in Q = \{1, \ldots, 96\}$) in the Day-Ahead Auction (DAA). State-of-charge (SOC) is tracked with $Q^+ = \{1, \ldots, 97\}$.

The dataset covers 2024–2025 and includes a market structure change: before 1 October 2025, DAA prices are hourly and trades must be placed as hourly blocks (four equal quarters per hour). From 1 October 2025 onward, DAA prices and trades are quarter-hourly, removing the hourly coupling constraint.

This calculation is an adaptation of the FlexPower BESS Optimizer / FlexIndex model. Many thanks to the FlexPower team for open-sourcing their contribution to the BESS optimization sector!

This is a simple model of flexibility asset optimization, which is aimed at transparency and comprehensibility. Therefore, more complex system parameters like efficiency or grid connection constraints are omitted.

The analysis excludes bid/ask acceptance effects, pay-as-cleared effects, and dynamic effects on bid and offer curves of the suboptimal trading decisions made on inaccurate price forecasts. As such, this analysis is meant as an initial, rough estimate of the impact of forecast accuracy on trading margin.

This document is meant to accompany the Python implementation of the optimization available on our GitHub.

Symbols

Sets and indices

Symbol	Meaning
$D = \{1, \ldots, 366\}$	day index (dataset covering 2024–2025)
$H = \{0, \ldots, 23\}$	hour index (used for hourly coupling constraint in pre-Oct 2025 period)
$Q = \{1, \ldots, 96\}$	quarter-hour index within the day
$Q^+ = \{1, \ldots, 97\}$	quarter-hour index for SOC including terminal state
$\mathcal{K} = \{\mathrm{P}, 1, \ldots, k, \mathrm{PF}\}$	price forecast index (Persistence, forecasts 1...k, Perfect Foresight)
$\hat{\lambda}^{(\mathrm{P})}$	Persistence forecast, where $\hat{\lambda}^{(\mathrm{P})}_{d,q} := \lambda_{d-1,q}$
$\hat{\lambda}^{(\mathrm{PF})}$	Perfect Foresight forecast, where $\hat{\lambda}^{(\mathrm{PF})}_{d,q} := \lambda_{d,q}$

Parameters

Symbol	Meaning
$\texttt{cap}^{\text{energy}}$	energy capacity [MWh]
$\texttt{cap}^{\text{power}}$	charge/discharge power limit [MW]
n_{cycles}	allowed daily equivalent full cycles [–]
$L_d = \texttt{cap}^{\text{energy}} \cdot n_{\text{cycles}}$	daily energy-throughput limit [MWh]
$\hat{\lambda}^{\text{DAA, k}}_{d,q}$	DAA forecast price from forecast k for d, q [EUR/MWh]
$\lambda^{\text{DAA}}_{d,q}$	DAA actual price for d, q [EUR/MWh]
$\Delta = \frac{1}{4}$	quarter-hour duration [h]

Decision variables

Symbol	Meaning
$\text{CHA}^{\text{DAA}}_{q,k},\ \text{DIS}^{\text{DAA}}_{q,k} \in [0,1]$	normalized charge/discharge in DAA at q, k
$\text{SOC}_{q,k}$	state-of-charge at $q, k \in Q^+$ [MWh]

Outputs

Symbol	Meaning
V_k	Value of forecast k compared with persistence forecast P
C_k	Cost of the error of forecast k compared with perfect foresight PF
$AR_{\text{total},k}$	Annual trading margin from DAA across all days $d \in D$, given forecast k
$DR_{d,k}$	Daily trading margin from DAA for day d, given forecast k

2 Calculating Outputs

2.1 Forecast Value

The value (per MW traded BESS) of a price forecast equals the difference in the annual DAA trading margin of trading decisions based on that forecast, as compared with the annual margin of trading decisions based on the persistence forecast.

$$V_k = AR_{\text{total},k} - AR_{\text{total},P} \tag{2.1.1}$$

2.2 Cost of Forecast Error

The cost (per MW traded BESS) of the error in a price forecast equals the difference in the annual DAA trading margin of trading decisions based on perfect foresight, as compared with the annual margin of trading decisions based on forecast k.

$$C_k = AR_{\text{total},PF} - AR_{\text{total},k} \tag{2.2.1}$$

2.3 Annual and Daily Trading Margin

The annual trading margin equals the sum of daily DAA trading margins across all days, given forecast k:

$$AR_{\text{total},k} = \sum_{d \in D} DR_{d,k} \tag{2.3.1}$$

The daily trading margin equals the sum of arbitrage margins across all quarters, evaluated at actual prices, given the trading decisions from the DAA optimization under forecast k:

$$DR_{d,k} = \sum_{q \in Q} \lambda^{\text{DAA}}_{d,q} \cdot \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\text{DIS}^{\text{DAA}}_{q,d,k} - \text{CHA}^{\text{DAA}}_{q,d,k}\right) \tag{2.3.2}$$

The factor $\texttt{cap}^{\text{power}} \cdot \Delta$ converts normalized charge/discharge values to MWh per quarter-hour.

3 Optimizing Daily Trading Decisions

3.1 Day-Ahead Auction (DAA)

Objective

The objective is to find the charge/discharge decisions for each quarter of day $d \in D$ which generate the highest expected revenue. Decisions are taken based on DAA forecast prices from forecast $k \in \mathcal{K}$.

$$\max_{\text{CHA}^{\text{DAA}}_{q,k}, \text{DIS}^{\text{DAA}}_{q,k}} \sum_{q \in Q} \hat{\lambda}^{\text{DAA}}_{q,k} \cdot \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\text{DIS}^{\text{DAA}}_{q,k} - \text{CHA}^{\text{DAA}}_{q,k}\right) \tag{3.1.1}$$

Constraints

State of charge in any quarter cannot exceed the maximum capacity ($\texttt{cap}^{\text{energy}}$):

$$\text{SOC}_q \leq \texttt{cap}^{\text{energy}} \quad \forall q \in Q^+ \tag{3.1.2}$$

State of charge in any quarter cannot be less than 0:

$$\text{SOC}_q \geq 0 \quad \forall q \in Q^+ \tag{3.1.3}$$

State of charge at the starting quarter is fixed as 0:

$$\text{SOC}_1 = 0 \tag{3.1.4}$$

State of charge at the terminal quarter is fixed as 0. Note: the terminal quarter is $q = 97$, not $q = 96$:

$$\text{SOC}_{97} = 0 \tag{3.1.5}$$

Step constraint: the state of charge in each quarter equals the previous quarter's SOC plus charging minus discharging energy:

$$\text{SOC}_{q+1} = \text{SOC}_q + \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\text{CHA}^{\text{DAA}}_q - \text{DIS}^{\text{DAA}}_q\right) \quad \forall q \in Q \tag{3.1.6}$$

Total physical charging must not exceed the daily energy-throughput limit:

$$\texttt{cap}^{\text{power}} \cdot \Delta \cdot \sum_{q \in Q} \text{CHA}^{\text{DAA}}_q \leq L_d \tag{3.1.7}$$

Total physical discharging must not exceed the daily energy-throughput limit:

$$\texttt{cap}^{\text{power}} \cdot \Delta \cdot \sum_{q \in Q} \text{DIS}^{\text{DAA}}_q \leq L_d \tag{3.1.8}$$

Hourly coupling constraint (pre-October 2025 only)

Before 1 October 2025, DAA prices are published and traded as hourly products. Trades must therefore be placed as equal blocks across all four quarters of each hour. For charging:

$$\text{CHA}^{\text{DAA}}_{4h+1} = \text{CHA}^{\text{DAA}}_{4h+2} = \text{CHA}^{\text{DAA}}_{4h+3} = \text{CHA}^{\text{DAA}}_{4h+4} \quad \forall h \in H \tag{3.1.9}$$

For discharging:

$$\text{DIS}^{\text{DAA}}_{4h+1} = \text{DIS}^{\text{DAA}}_{4h+2} = \text{DIS}^{\text{DAA}}_{4h+3} = \text{DIS}^{\text{DAA}}_{4h+4} \quad \forall h \in H \tag{3.1.10}$$

From 1 October 2025 onward, DAA trades are quarter-hourly and these coupling constraints are dropped.